SYNOPSIS

On

An AI-Based Framework for Pre-Harvest Crop Yield and Production Estimation

Group ID:_____

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1. Title of the Project

An AI-Based Framework for Pre-Harvest Crop Yield and Production Estimation

2. Abstract

Accurate pre-harvest estimation of crop yield and total production is an important requirement in orchard and precision-agriculture management. Conventional estimation commonly depends on manual sampling, visual judgement and extrapolation from a limited number of trees or field locations, which can be labour-intensive and may not adequately represent spatial variation within an orchard. Recent research has demonstrated that computer vision can detect fruits, estimate crop stage and physical size, and support yield-related estimation, but these capabilities are often studied as separate tasks. The proposed M.Tech project investigates an AI-based framework that converts pre-harvest field observations into an estimate of expected crop yield and total production. The initial implementation will use pomegranate as the case study, while the system architecture will be designed so that the methodology can later be adapted to other crops. The proposed pipeline will combine orchard metadata with RGB images or video, perform fruit detection or segmentation, estimate fruit count and selected physical attributes, correct for visibility and occlusion effects where required, and transform tree-level observations into yield estimates and orchard-level production estimates. Machine-learning regression and calibration methods will be evaluated against appropriate baselines. The study will emphasize quantitative validation using ground-truth harvest measurements and metrics such as counting error, MAE, RMSE, MAPE and R^2. The final outcome will be a research-oriented Python-based prototype that reports estimated yield per tree or unit area and expected total production for the assessed orchard, together with measurable uncertainty or error information. The project is intentionally restricted to yield and production estimation; price forecasting, monetary valuation, contract pricing and trader-economics modelling are outside the current scope.

3. Introduction / Background

Yield estimation is a core precision-agriculture task because growers need an early estimate of expected production for harvest preparation, labour planning, storage, transportation and resource management. Manual yield estimation typically requires field sampling and later extrapolation to a larger orchard or production block. Such sampling can be costly, time-consuming and sensitive to sampling density and spatial variability.

Computer vision provides a non-destructive mechanism for collecting dense observations of standing crops. Existing research has shown that image segmentation and fruit detection can be used to count fruit and estimate orchard-level yield, while other studies demonstrate pomegranate growth-stage recognition, lightweight fruit detection, on-tree monitoring and physical fruit-size estimation [1–5]. However, these studies address different parts of the estimation chain and do not by themselves provide a complete, crop-specific pre-harvest production-estimation workflow.

The proposed work therefore focuses on a narrower and measurable research problem: estimating expected yield from observable pre-harvest crop characteristics and converting that estimate into total production for an orchard or defined production area. Pomegranate is selected as the initial case study because several recent studies provide a relevant technical foundation for pomegranate image analysis, while the developed architecture will be kept modular for later crop extension [2–5].

4. Problem Definition

Pre-harvest crop yield is difficult to estimate accurately because only part of the crop may be directly visible in images, fruit may be clustered or occluded by leaves and branches, and fruit number alone does not directly

provide harvested mass. Manual sampling also provides sparse observations that must be extrapolated to the full orchard. There is therefore a need for a validated AI-based framework that combines visual crop observations with orchard-level information to estimate tree-level yield and aggregate those estimates into expected total production before harvest.

Research focus

The project estimates physical crop output, not monetary crop value. The primary outputs are expected yield per tree/unit area and expected total production for the assessed orchard or crop block.

5. Objectives

1. To develop a reproducible dataset and field-data protocol for pre-harvest pomegranate yield estimation using RGB images/video and orchard metadata.
2. To design a computer-vision pipeline for fruit detection or segmentation and crop-load estimation under realistic orchard conditions, including occlusion and illumination variation.
3. To estimate selected fruit-level attributes, such as apparent size or estimated mass, where these attributes improve conversion from fruit counts to yield in kilograms.
4. To develop and compare machine-learning models for tree-level or unit-area yield estimation using visual features and relevant orchard metadata.
5. To aggregate tree-level or sampled-unit estimates into an orchard-level production estimate and quantify the resulting estimation error.
6. To evaluate the proposed framework against suitable baselines using counting error, MAE, RMSE, MAPE and R^2, together with calibration/uncertainty analysis where supported by the available ground truth.

6. Scope of the Project

- Initial crop: pomegranate; later adaptation to other crops will be considered only after the core pipeline is validated.
- Primary task: pre-harvest yield estimation and total production estimation.
- Input data: RGB images/video of standing crop, orchard metadata, tree/plant information and available field measurements.
- Computer vision: fruit detection/segmentation, counting, crop-load features and selected size/weight estimation where useful.
- Prediction: tree-level yield, unit-area yield and/or orchard-level production in physical units such as kg or tonnes.
- Validation: comparison with actual harvested production or reliable field ground truth collected using a defined sampling protocol.
- Research experiments: model comparison, feature ablation, sampling-density analysis, robustness to occlusion/illumination and generalization checks where data permit.
- Out of scope: commodity-price forecasting, crop valuation in rupees, farmer-trader contract pricing, trader margin estimation and supply-chain profit optimization.

7. Research Positioning and Literature-Derived Gap

The reviewed literature provides complementary evidence rather than a single complete solution:

[1] Bargoti and Underwood demonstrated fruit segmentation, detection and row-level yield estimation in an apple orchard, including calibration against post-harvest counts. The study shows that image-derived fruit counts can support yield estimation but also highlights the need for calibration and careful handling of occlusion.

[2] Naseer et al. developed a transfer-learning approach for pomegranate growth-stage classification. This provides crop-state information but does not estimate orchard production.

[3] Zhao et al. developed a lightweight pomegranate fruit detector for mobile/edge deployment. Detection performance supports crop-load measurement, but detection alone does not provide calibrated mass yield or total orchard production.

[4] Giménez-Gallego et al. used repeated on-tree observations for fruit monitoring and harvest-date prediction. The work demonstrates temporal crop monitoring but its primary output is harvest timing rather than total production.

[5] Devanna et al. investigated RGB-D based pomegranate size estimation. Physical size can support yield conversion, but the study does not establish an orchard-scale production estimator.

Research gap

The literature reviewed supports individual components of yield estimation—fruit detection, crop-stage assessment, temporal monitoring and physical sizing—but does not establish a unified, validated pre-harvest framework that combines these observations with orchard-level calibration to estimate both yield and total production for a standing pomegranate orchard. The proposed work investigates this integration and evaluates its accuracy under realistic field conditions.

8. Proposed System

The proposed system will transform pre-harvest field observations into quantitative yield and production estimates through the following architecture:

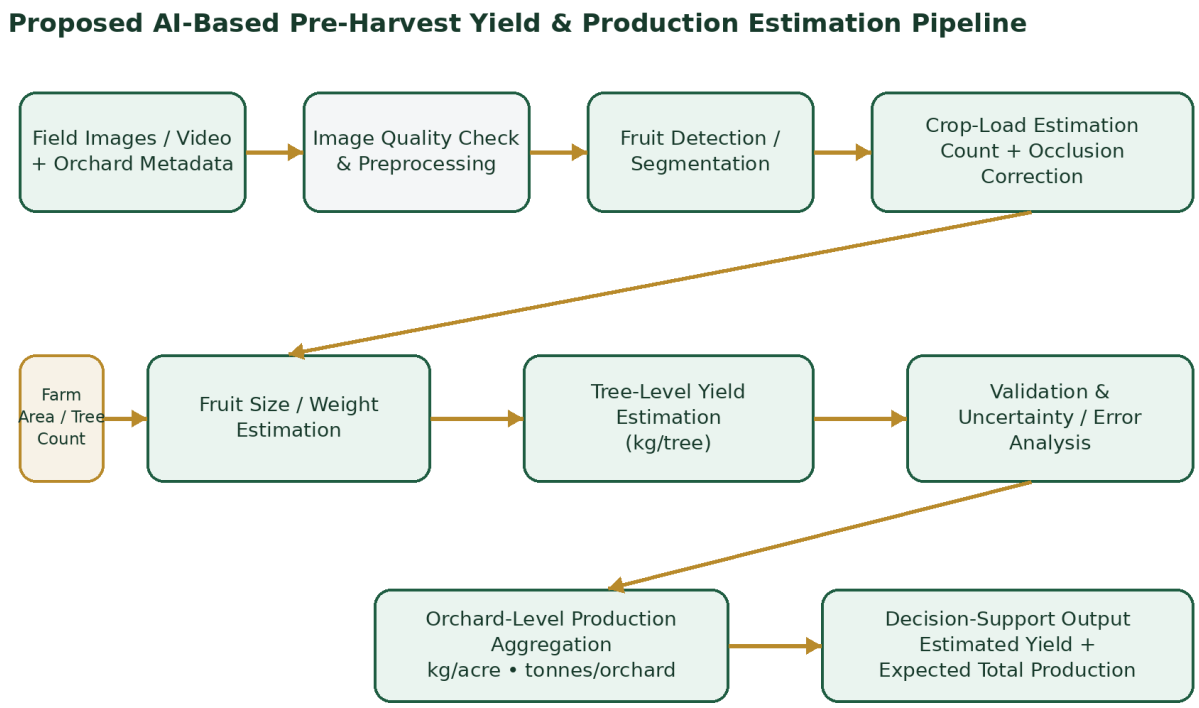


Figure 1. Proposed AI-based pre-harvest yield and production estimation architecture

Main Modules

1. Data Acquisition & Metadata: Capture orchard images/video together with tree/row/area information, image location or sampling identifiers and available field measurements.

2. Image Quality & Preprocessing: Normalize image format and quality, resize where required, and prepare data for model inference while preserving spatial sampling information.

3. Fruit Detection / Segmentation: Detect or segment visible fruits using an appropriate deep-learning model selected through baseline comparison and validation.

4. Crop-Load & Visibility Estimation: Convert detections into crop-load indicators and, where required, estimate missing/occluded fruit contribution using calibration or learned correction.

5. Fruit Size / Mass Estimation: Estimate apparent fruit size and investigate size-to-weight relationships when they improve conversion from fruit count to physical yield.

6. Tree-Level Yield Estimator: Combine visual crop-load features with relevant orchard metadata using regression or other supervised models to estimate yield per tree/unit area.

7. Orchard Production Aggregator: Aggregate validated tree or sampling-unit estimates using orchard area/tree count and sampling design to obtain total expected production.

8. Validation & Uncertainty: Compare predictions with ground truth, quantify errors, and report an estimation range or uncertainty indicator when statistically supported.

9. Research Prototype: Provide a simple Python-based interface/report that presents estimated yield, total production and validation diagnostics for an assessed orchard.

9. System Requirements

a. Hardware Requirements

- Intel Core i5 / AMD Ryzen 5 or equivalent processor or better.
- 16 GB RAM recommended; 8 GB minimum for lighter experiments.
- GPU with at least 6 GB VRAM recommended for deep-learning training; CPU-only development is possible for preprocessing and smaller experiments.
- At least 50 GB free storage for datasets, model checkpoints, experiments and project documentation.
- Camera-enabled mobile device, RGB camera or suitable field imaging setup for data acquisition.
- Stable internet connection for literature, dataset management and optional cloud compute.

b. Software Requirements

- Python 3.11/3.12.
- Antigravity IDE or VS Code; Git and GitHub for version control.
- OpenCV, NumPy, Pandas and Matplotlib for image/data processing.
- PyTorch and/or an appropriate YOLO/segmentation framework for computer-vision experiments.
- scikit-learn for regression, baselines and evaluation.
- Jupyter Notebook for controlled research experiments and visualization.
- Optional Streamlit/FastAPI prototype layer for field-assessment demonstration.

10. Methodology / Implementation Plan

Phase 1 — Problem and Literature Finalization: Finalize the yield/production definition, case-study crop, measurable research questions, ground-truth protocol and evaluation metrics.

Phase 2 — Data Acquisition Protocol: Define orchard sampling strategy, imaging positions, image/video capture conditions and associated metadata. Collect representative examples covering illumination, canopy density, fruit clustering and maturity variation.

Phase 3 — Dataset Preparation: Annotate fruit instances or masks, organize train/validation/test partitions, document class/scene distribution and apply leakage-safe preprocessing and augmentation.

Phase 4 — Fruit Detection / Segmentation Baselines: Evaluate suitable object-detection or segmentation approaches and establish a reproducible baseline for fruit counting under field conditions.

Phase 5 — Crop-Load and Size Features: Generate count, density, visible-fruit ratio and selected size-related features. Investigate methods to reduce double counting and account for partial visibility.

Phase 6 — Yield Estimation Model: Train and compare regression models using image-derived features and orchard metadata. Establish simple baselines before applying more complex models.

Phase 7 — Production Aggregation: Convert tree-level or sampled-unit estimates into unit-area and whole-orchard production using the defined sampling/calibration strategy.

Phase 8 — Experimental Evaluation: Measure detection/counting metrics and production-estimation metrics. Conduct feature ablation, sampling-density analysis and robustness experiments where data permit.

Phase 9 — Ground-Truth Validation: Compare predictions with actual harvested quantities or carefully collected field measurements and quantify systematic bias and error.

Phase 10 — Prototype and Reporting: Package the validated pipeline into a Python-based prototype, document limitations and prepare dissertation-ready experimental results.

Evaluation Framework

Component	Primary Metrics	Purpose
Fruit detection / segmentation	Precision, Recall, F1, mAP, counting error	Measure reliability of crop-load observations.
Fruit/weight estimation	MAE, RMSE, MAPE, R^2	Quantify physical-yield conversion error.
Tree / area yield	MAE, RMSE, MAPE, R^2	Evaluate predicted kg/tree or kg/unit area.
Orchard production	Absolute error, percentage error, bias	Measure difference between estimated and actual total production.
Robustness	Performance under occlusion, illumination and sampling changes	Assess whether results remain stable outside ideal conditions.

11. Expected Outcome

- A reproducible Python-based computer-vision pipeline for pre-harvest crop-load observation and yield estimation.
- A documented pomegranate orchard image dataset and annotation/data dictionary prepared according to the selected experimental protocol.
- A validated fruit detection/segmentation model or model comparison with clearly reported performance.
- A calibrated method for converting image-derived crop-load information into tree-level or unit-area yield estimates.
- An orchard-level production estimation procedure that reports expected total production in kg/tonnes together with measured error.
- Experimental evidence showing the contribution of fruit count, size/weight and orchard metadata through baseline and ablation comparisons.
- A Python-based research prototype that produces a structured yield and production assessment report.
- A documented set of limitations, failure cases and conditions under which the estimator should or should not be trusted.

12. Applications / Use Cases

- Pre-harvest harvest planning: support early estimation of expected crop volume for labour and logistics planning.
- Orchard management: provide spatial or tree-level yield information for identifying variability within production blocks.
- Storage and transportation planning: support estimation of expected physical produce volume before harvest.
- Yield monitoring: enable repeated assessments during the season to study crop-load changes.
- Precision agriculture research: provide a computer-vision and ML framework that can later be adapted to other crops.

13. Limitations and Research Risks

- Occlusion and overlapping fruit may cause under-counting; correction methods must be validated against actual field conditions.
- Changes in illumination, camera viewpoint, canopy structure and fruit appearance can reduce model generalization.
- Image-derived fruit counts are not automatically equivalent to harvested kilograms; a calibration relationship is required.
- Ground-truth harvest data are essential for credible production validation and may be expensive to collect.
- A model trained on one orchard, cultivar or season may not generalize to other orchards without additional data.
- The project will not interpret an estimate as an exact harvest quantity; error bounds and data limitations must accompany the result.

14. Future Enhancements / Future Scope

- Extend the validated pipeline from pomegranate to additional crops with crop-specific detection and calibration modules.
- Integrate multi-view or temporal image sequences to reduce occlusion-related under-counting and improve crop-load estimation.
- Investigate RGB-D or other depth-based sensing when field cost and data availability justify the added complexity.
- Develop spatial yield maps using geotagged observations for orchard-level variability analysis.
- Incorporate weather, crop-management and historical yield data as additional predictors for improved yield forecasting.
- Evaluate probabilistic or quantile-regression methods for calibrated prediction intervals rather than only point estimates.
- Study transfer learning and domain adaptation for new orchards, cultivars and seasons.
- Integrate the production-estimation output with downstream planning systems; monetary valuation and market-price modules are explicitly outside the current M.Tech scope.

15. Project Deliverables

- Annotated and documented research dataset for the selected crop and field conditions.
- Computer-vision model(s) for fruit detection/segmentation and crop-load measurement.
- Yield-estimation and production-aggregation models with saved experiment configurations.
- Evaluation report containing ground-truth comparison, errors, robustness studies and ablation results.
- Python-based prototype for entering/processing orchard observations and presenting estimated yield/production.
- M.Tech dissertation chapters, technical documentation and research-paper-ready experimental results.

16. References

[1] S. Bargoti and J. P. Underwood, “Image Segmentation for Fruit Detection and Yield Estimation in Apple Orchards,” arXiv:1610.08120, 2016. (The supplied paper reports CNN/MLP segmentation, fruit detection and orchard-row yield estimation.)

[2] Z. Naseer et al., "A Novel Transfer Learning Approach for Detection of Pomegranates Growth Stages," IEEE Access, vol. 12, pp. 27073–27087, 2024. DOI: 10.1109/ACCESS.2024.3365356.

[3] Y. Zhao et al., "YOLO-Granada: A Lightweight Attentioned Yolo for Pomegranates Fruit Detection," Scientific Reports, vol. 14, 16848, 2024. DOI: 10.1038/s41598-024-67526-4.

[4] J. Giménez-Gallego et al., "Fruit Monitoring and Harvest Date Prediction Using On-Tree Automatic Image Tracking," IEEE Transactions on AgriFood Electronics, vol. 3, no. 1, pp. 56–68, 2025. DOI: 10.1109/TAFE.2024.3408912.

[5] R. P. Devanna et al., "Automated On-Tree Detection and Size Estimation of Pomegranates by a Farmer Robot," Robotics, vol. 14, no. 10, Art. 131, 2025. DOI: 10.3390/robotics14100131.

Scope note: This synopsis intentionally excludes monetary valuation, market-price forecasting, contract pricing and trader-margin estimation. The research contribution is centered on physical pre-harvest yield and total production estimatio